

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

DIGITAL IMAGE PROCESSING

(Electronics and Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) Mention the role of image sampling in digital image processing. 2M
 - (b) Compare Discrete Cosine Transform (DCT) and Walsh Transform. 2M
 - (c) What problem would arise if an image histogram is heavily skewed towards lower intensity values, and how could basic intensity transformations help correct this? 2M
 - (d) If an image has both high noise and low contrast, Outline the sequence of intensity transformations and filtering methods to improve the image. 2M
 - (e) List the two approaches for blind image restoration. 2M
 - (f) Mention the steps involved in frequency domain filtering. 2M
 - (g) Why is LZW coding preferred for text-based image compression over other methods, such as Huffman coding? 2M
 - (h) Outline the role of wavelet transforms in image compression. 2M
 - (i) How is edge detection used to detect discontinuities in a digital image? 2M
 - (j) How does thresholding differ from region-based segmentation? 2M

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Describe the fundamental steps in digital image processing. Also, Summarize the basic relationships between pixels in a digital image. 10M

OR

- 3 With Discrete Fourier Transform (DFT) explain how an image is transformed from the spatial domain to the frequency domain. Provide a step-by-step calculation for a small image matrix. 10M

- 4 You are tasked with sharpening an image that contains significant background noise. Summarize the steps to apply both spatial and frequency domain sharpening filters. Interpret the combination of filters would you choose to maintain edge detail while minimizing noise, and Justify. 10M

OR

- 5 An image has low contrast and poor visibility of details. Explain a sequence of intensity transformations and histogram processing methods that can enhance the image. Analyze each step in improving the image quality. 10M

- 6 Describe the key components of the image degradation/restoration process. Interpret the role the degradation function plays in this model. 10M

OR

- 7 Explain the working of Wiener filtering in the presence of additive Gaussian noise. How does it achieve minimum mean square error in the restoration process? 10M

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- 8 Solve and find a Huffman code and the average length of the code and its redundancy for the source emits letters from an alphabet $A = \{a_1, a_2, a_3, a_4, a_5\}$ with probabilities $P(a_1) = 0.2$, $P(a_2) = 0.4$, $P(a_3) = 0.2$, $P(a_4) = 0.1$ and $P(a_5) = 0.1$. 10M

OR

- 9 Analyze the effectiveness of Run-Length coding for compressing a simple binary image compared to Bit-Plane coding. 10M

- 10 Evaluate the effectiveness of watershed segmentation in morphological image processing. Also, explain pre-processing techniques like noise removal and edge detection to improve its performance. 10M

OR

- 11 Explain the significance of morphological image processing in boundary extraction. Describe how dilation and erosion can be combined to extract boundaries from a binary image. 10M
